V k - sp. vect.

$U_1, U_2 \subseteq V$ subspații vect. $\longrightarrow$ $U_1 \cap U_2$, $U_1 \cup U_2$

$U_1 \cap U_2$ subspațiu ? DA

$\forall \lambda, \mu \in V \quad \forall v, w \in U_1 \cap U_2 \Rightarrow \lambda v + \mu w \in U_1 \cap U_2$

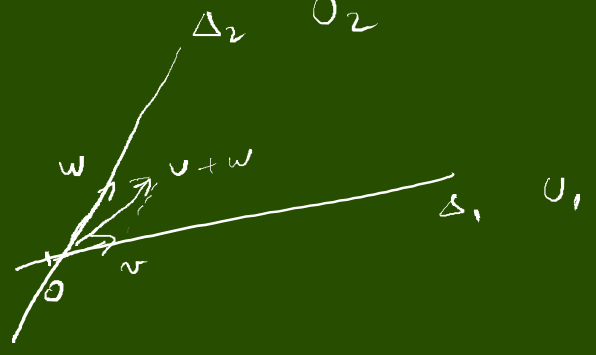
$v, w \in U_1 \cap U_2 \Rightarrow v, w \in U_1 \Rightarrow \lambda v + \mu w \in U_1$

$v, w \in U_2 \Rightarrow \lambda v + \mu w \in U_2$

$\Rightarrow \lambda v + \mu w \in U_1 \cap U_2$

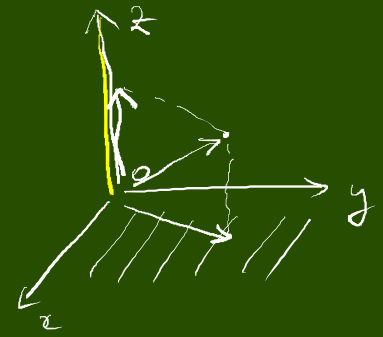
$U_1 \cup U_2$ subspațiu ? NU

R^2



Exemple R^3

$U_1 = \{ \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} | x, y \in R \} = \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} | z = 0 \}$ xOy



$U_2 = \{ \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix} | z \in R \}$ Oz

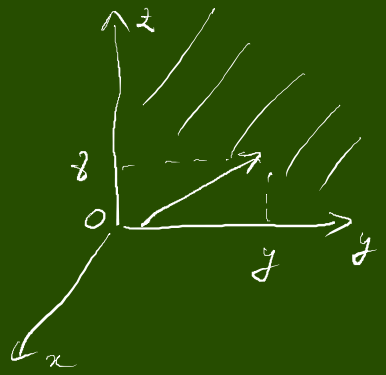
$= \{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} | x = 0, y = 0 \}$

$U_1 + U_2 = R^3$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix}$

$\in U_1 \quad \in U_2$

$U_3 = \{ \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix} | y \in R \}$ Oy



$U_2 + U_3 =$

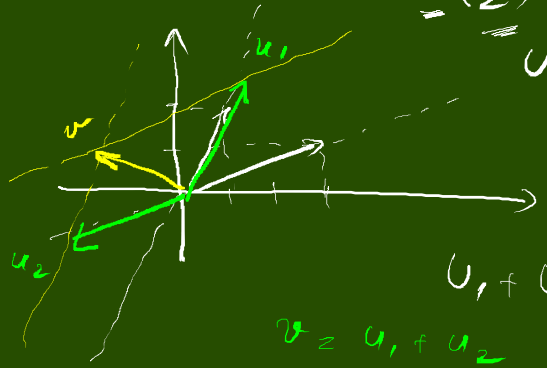
$y\mathbb{R}^2: \begin{pmatrix} 0 \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ y \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ z \end{pmatrix}$

$\in U_3 \quad \in U_2$

$\mathbb{R}^2$

$U_1 = \left\{ \begin{pmatrix} x \\ 2x \end{pmatrix} \mid x \in \mathbb{R} \right\}$

$= x \begin{pmatrix} 1 \\ 2 \end{pmatrix}$



$U_2 = \left\{ \begin{pmatrix} 3y \\ y \end{pmatrix} \mid y \in \mathbb{R} \right\}$

$y \begin{pmatrix} 3 \\ 1 \end{pmatrix}$

$U_1 + U_2 = ?$

$v = u_1 + u_2$

$u_1 \in U_1, \quad u_2 \in U_2$

$v = \begin{pmatrix} x \\ 2x \end{pmatrix} + \begin{pmatrix} 3y \\ y \end{pmatrix} \Rightarrow \begin{matrix} x = \dots \\ y = \dots \end{matrix}$

$\cdots = x + 3y$

$\cdots = 2x + y$

$\begin{pmatrix} 1 & 3 \\ 2 & 1 \end{pmatrix}$

$A \in \mathcal{M}_{m,n}(k)$

$\ker(A) \subseteq k^n$ $\qquad$ $\operatorname{Im}(A) \subseteq k^m$

$Ax = O_m$ $\qquad$ b $\qquad$ $Ax = b$

Exemplu: $A = \begin{pmatrix} 1 & -3 & 2 \\ 2 & 0 & 1 \\ 3 & -3 & 3 \end{pmatrix}$

$\ker(A) = ?$ $\qquad$ $\operatorname{Im}(A) = ?$

$\ker(A) \subseteq \mathbb{R}^3$ $\quad 3 = nr.$ coloane

$\ker(A) = \{ X \in \mathbb{R}^3 \mid AX = O_3 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \}$

$X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$AX = O_3 \Rightarrow \begin{cases} x - 3y + 2z = 0 \\ 2x + z = 0 \\ 3x - 3y + 3z = 0 \end{cases}$$

$z = -2x \Rightarrow y = -x$

$$\Rightarrow \underline{Ker(A)} = \left\{ \begin{pmatrix} x \\ -x \\ -2x \end{pmatrix} \mid x \in \mathbb{R} \right\}$$

k^2

$$= \left\{ x \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \mid x \in \mathbb{R} \right\} = Sp\left\{ \begin{pmatrix} 1 \\ -1 \\ -2 \end{pmatrix} \right\}$$

3D dreaptă cu bază pun 0

$$Im(A) = \left\{ B = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{R}^3 \mid AX = B \text{ compat.} \right\}$$

$$\begin{cases} x - 3y + 2z = a \\ 2x + z = b \\ 3x - 3y + 3z = c \end{cases}$$

$$\begin{vmatrix} 1 & -3 & a \\ 2 & 0 & b \\ 3 & -3 & c \end{vmatrix} = 0 \Rightarrow a + b = c$$

$$Im(A) = \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{R}^3 \mid a + b = c \right\}$$

$0 \in$ plan

$$= \left\{ \begin{pmatrix} a \\ b \\ a+b \end{pmatrix} \mid a, b \in \mathbb{R} \right\}$$

$$= \left\{ a \cdot \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + b \cdot \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \mid a, b \in \mathbb{R} \right\} = \mathrm{Sp}\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\}$$

Exemple. 1) RGB $\begin{pmatrix} r \\ g \\ b \end{pmatrix} =$

$$= \frac{r}{255} \begin{pmatrix} 255 \\ 0 \\ 0 \end{pmatrix} + \frac{g}{255} \begin{pmatrix} 0 \\ 255 \\ 0 \end{pmatrix} + \frac{b}{255} \begin{pmatrix} 0 \\ 0 \\ 255 \end{pmatrix}$$

2) QC

Soit $\psi = \alpha |0\rangle + \beta |1\rangle$

$\alpha, \beta \in \mathbb{C}$

$|\alpha|^2 + |\beta|^2 = 1$

Dirac (ket) ↑ Bloch

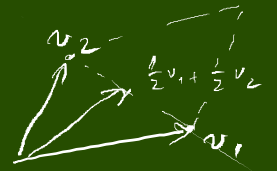
3) $\mathbb{R}^2$ $\quad v_1 \quad v_2 \quad \frac{1}{2}v_1 + \frac{1}{2}v_2$

$\lambda_1 = \lambda_2 = \frac{1}{2}$

Convessa $\quad \alpha v_1 + (1-\alpha)v_2$

$\alpha \in (0,1)$

$\alpha \in \mathbb{R}$



$V = Sp\{v_1, \dots v_n\}$ Span

Sist. di generatori per V

$V = k[x]$ $\quad 1, x, x^2, x^3, \dots, x^n, \dots$

$Sp\{1, x, x^2, \dots, x^n, \dots\} = k[x]$

$a_0 1 + a_1 x + \dots + a_n x^n$

$= \quad = \quad =$

$$\lambda_1 v_1 + \dots + \lambda_n v_n + \lambda_{n+1}(v_{n-1} + v_n) =$$

$$= \lambda_1 v_1 + \dots + (\lambda_{n-1} + \lambda_{n+1}) v_{n-1} + (\lambda_n + \lambda_{n+1}) v_n$$

$$\in Sp\{v_1 \dots v_n\}$$

$$A = \begin{pmatrix} 1 & -3 & 2 \\ 2 & 0 & 1 \\ \underbrace{3}_{v_1} & \underbrace{-3}_{v_2} & \underbrace{3}_{v_3} \end{pmatrix}$$

$$Im(A) = \{ B \in \mathbb{R}^3 \,/\, \exists X \in \mathbb{R}^3 . \; AX = B \}$$

B come $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ — X come $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$

$$\begin{cases} x - 3y + 2z = a \\ 2x + z = b \\ 3x - 3y + 3z = c \end{cases}$$

$$x \underbrace{\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}}_{v_1} + y \underbrace{\begin{pmatrix} -3 \\ 0 \\ -3 \end{pmatrix}}_{v_2} + z \underbrace{\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}}_{v_3} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

$$\mathrm{Im}(A) = \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \mid \exists\, x, y, z \in \mathbb{R} \text{ s.t. } \right.$$

$$\left. \begin{pmatrix} a \\ b \\ c \end{pmatrix} = x \cdot v_1 + y \cdot v_2 + z \cdot v_3 \right\}$$

$$= \mathrm{Sp}\{ v_1, v_2, v_3 \}$$

$A \in M_{m,n}(k)$ $\quad \mathrm{Im}(A) \subseteq k^m$

$\mathrm{Im}(A) = k^m$?

A? s.t. $AX = B$ compatibile $\forall B$?

$$\begin{cases} a_{11} x_1 + \dots = b_1 \\ \dots \\ a_{m1} x_1 + \dots + a_{mn} x_n = b_m \end{cases}$$

$\mathrm{rang}(A) = m$ $\quad$ ($m \leq n$)

$m = n$

Exemplu: $\mathbb{R}^2$ $v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $v_2 = \begin{pmatrix} -3 \\ 2 \end{pmatrix}$

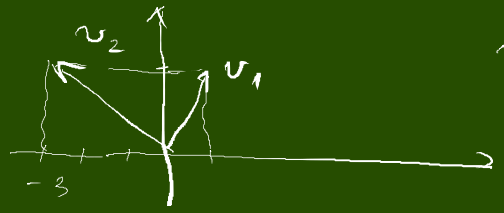
$\begin{cases} \text{liniar independ.} ? \checkmark \\ \text{liniar dep.} ? \end{cases}$

Fie $\lambda_1, \lambda_2 \in \mathbb{R}$ a.i. $\lambda_1 v_1 + \lambda_2 v_2 = O_2 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\lambda_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$$\begin{cases} \lambda_1 - 3\lambda_2 = 0 \\ 2\lambda_1 + 2\lambda_2 = 0 \end{cases} \Rightarrow \begin{matrix} \lambda_2 = 0 \\ \lambda_1 = 0 \end{matrix}$$

$\Rightarrow v_1, v_2$ liniar independenți



$v_1 = \vec{i} + 2\vec{j}$ $v_2 = -3\vec{i} + 2\vec{j}$

$v_1 \cdot v_2 \neq 0$

$v_1 \not\perp v_2$

$\mathbb{R}^2 = \mathbb{R}^2$ $\quad v_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad v_3 = \begin{pmatrix} -2 \\ -4 \end{pmatrix}$

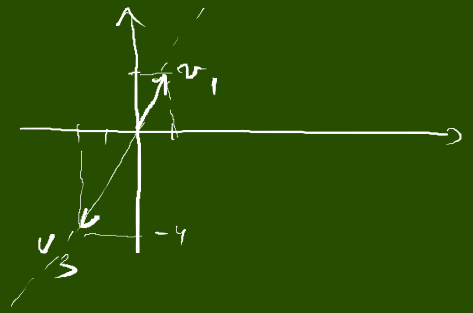
lin. indep. ?

Fie $\lambda_1, \lambda_3 \in \mathbb{R}$ a.i. $\lambda_1 v_1 + \lambda_3 v_3 = 0$

$$\begin{cases} \lambda_1 - 2\lambda_3 = 0 \\ 2\lambda_1 - 4\lambda_3 = 0 \end{cases} \Rightarrow \lambda_1 = 2\lambda_3$$

$\Rightarrow \boxed{2v_1 + v_3 = 0}$ lin. dep.

$\neq 0$



$\lambda \cdot v = 0$ $\begin{cases} \lambda = 0 \\ v = 0 \end{cases}$

$v_1 \dots v_n$ <u>lin. dep.</u>

$\lambda_1 v_1 + \dots + \lambda_n v_n = 0 \quad / \cdot \lambda_1^{-1}$

$\lambda_1 \in K^* = K \setminus \{0\}$

$v_1 = -\lambda_2 \lambda_1^{-1} v_2 + \dots + (-\lambda_n \lambda_1^{-1}) v_n$

<u>Exemplu</u> $v_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, v_2 = \begin{pmatrix} -3 \\ 0 \\ -3 \end{pmatrix}, v_3 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$

v_1, v_2, v_3 lin. indep ?

$\exists \lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ a.î. $\neq 0$ → cel puțin unul

$\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 = 0$

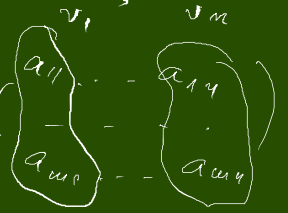
$$A = \begin{pmatrix} 1 & -3 & 2 \\ 2 & 0 & 1 \\ 3 & -3 & 3 \end{pmatrix}$$

$v_1 \quad v_2 \quad v_3$

$\operatorname{Ker} A = \operatorname{Sp}\{(\ )\} \neq \{0_3\}$

$\Leftrightarrow$ v_1, v_2, v_3 lin. dependent

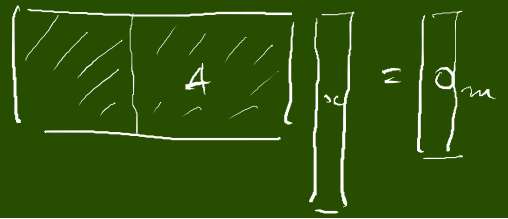
$A \in M_{m,n}(k)$ $\quad A = (\ a_{ij}\ )$



$A = (v_1 | \dots | v_n)$

$v_1 \dots v_n \in k^m$ lin. indep. $\Leftrightarrow$

~~$\operatorname{rang} A = m \leq n$~~



$\operatorname{rang} A = n \leq m$

$$A \begin{pmatrix} x \\ \vdots \end{pmatrix} = \begin{pmatrix} p \\ \vdots \end{pmatrix}_m$$

Tema $V = M_2(\mathbb{C})$ $\quad k = \mathbb{C}$

$A_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $A_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$, $A_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

Pauli

Q : A_1, A_2, A_3 liniar indep ?

$\lambda_1, \lambda_2, \lambda_3 \in \mathbb{C}$

$\lambda_1 A_1 + \lambda_2 A_2 + \lambda_3 A_3 = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$\overset{?}{\Rightarrow} \lambda_1 = \lambda_2 = \lambda_3 = 0$